1: An insurance company determines that N, the number of claims received in a week, satisfies $\Pr[N = n] = \frac{1}{2^{n+1}}$ where $n \geq 0$.

The company also determines that the number of claims received in a given week is independent of the number of claims received in any other week.

Determine the probability that exactly seven claims will be received during a two-week period.

A. $\frac{1}{256}$ B. $\frac{1}{128}$ C. $\frac{7}{512}$ **D.** $\frac{1}{64}$ E. $\frac{1}{32}$

Solution: Let N_1 and N_2 be the number of claims received in the first and second week respectively. Exactly seven claims are received if $N_1 = k$ and $N_2 = 7-k$ for some $k \in \{0, 1, \ldots, 7\}$, so by independence

$$\begin{aligned}
\Pr[N_1 + N_2 = 7] &= \sum_{k=0}^{7} \Pr[N_1 = k] \Pr[N_2 = 7 - k] \\
&= \sum_{k=0}^{7} \frac{1}{2^{k+1}} \cdot \frac{1}{2^{8-k}} \\
&= \sum_{k=0}^{7} \frac{1}{2^9} \\
&= \frac{8}{512} \\
&= \boxed{\frac{1}{64}}
\end{aligned}$$

and thus $\boxed{\text{D}}$ is the correct answer.

2: The ages of students enrolling in Mathematical Theory of Probability has a cumulative probability distribution given by:

x	17	18	19	20	21	22	23
$F(x)$	0	0.18	0.47	0.85	0.94	0.99	1.00

(a) Calculate the expected value for X

(b) Calculate the variance for X

(c) Calculate the standard deviation for X

Solution: We first recover the probability mass function using $p(x) = F(x) - F(x-1)$:

x	17	18	19	20	21	22	23
$p(x)$	0	0.18	0.29	0.38	0.09	0.05	0.01

(a)

$$\begin{aligned}
E[X] &= \sum_{x} x\, p(x) \\
&= 18(0.18) + 19(0.29) + 20(0.38) + 21(0.09) + 22(0.05) + 23(0.01) \\
&= 3.24 + 5.51 + 7.60 + 1.89 + 1.10 + 0.23 \\
&= \boxed{19.57}
\end{aligned}$$

(b) We first compute

$$\begin{aligned} E[X^2] &= \sum_x x^2\, p(x) \\ &= 324(0.18) + 361(0.29) + 400(0.38) + 441(0.09) + 484(0.05) + 529(0.01) \\ &= 58.32 + 104.69 + 152 + 39.69 + 24.2 + 5.29 \\ &= 384.19 \end{aligned}$$

so

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = 384.19 - 19.57^2 = 384.19 - 382.9849 = \boxed{1.2051}$$

(c)

$$\sigma_X = \sqrt{\mathrm{Var}(X)} = \sqrt{1.2051} \approx \boxed{1.0978}$$

[0] LaTeX code for this document can be found on github here